VINNA: Verifiable Rewards for Spatially-Anchored Construction Safety Incentives

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Abstract

Construction sites lose an estimated $177B annually to productivity gaps [16]; existing AI safety systems fail to close this gap because any classifier-driven reward signal is gameable — financially-motivated workers learn to trigger "productive" labels without performing real work. VINNA introduces an evidence-anchored reward ledger that binds every reward candidate to three independent technical witnesses — a COLMAP-derived camera pose from structure-from-motion reconstruction, a local VLM semantic object annotation, and a CII activity classification — gated by a four-condition verification protocol that blocks all payouts until human ground-truth review confirms the candidate window. On a full egocentric masonry video (1,276 s), our CII classifier assigns **86.7%** of 30 sampled frames as Productive (P) (n=30, 95% CI [69, 96], Clopper-Pearson) with mean confidence 0.939; spatially-anchored indexing from 31 uniformly sampled frames yields a sparse 3D map (1,770 points, mean reprojection error 1.199 px) that cryptographically links each reward candidate to a registered camera pose at the time of the productive event. We define a formal reward calculus: a binary OSHA-aligned per-frame reward $R_{\text{bin}}(f) \in \{0, 1\}$ and a continuous worker-level reward $R_{\text{SNRA}}(w) = \sigma\left(\sum_{f \in w} R_{\text{bin}}(f)\right)$, each gated by spatial anchor presence and human review. Spatially-anchored indexing is not an accuracy improvement over detection-only systems — it is a *prerequisite* for any construction reward system that must resist adversarial gaming and withstand contested audits.

Keywords

egocentric video, construction safety, verifiable rewards, structure from motion, spatially-anchored incentives, RLHF, agentic AI, Solana

1 Introduction

The construction industry is the largest employer of blue-collar labor in the United States, yet it remains one of the least data-instrumented sectors in the economy. OSHA estimates that construction accounts for 21% of all private-industry fatalities [15], despite representing roughly 6% of the workforce. The root cause is not lack of regulation but lack of measurement: without granular, machine-readable data about what workers are doing and where, both safety interventions and productivity incentives are administered at coarse-grained shift or day-level resolution.

Recent progress in egocentric video AI has created an opening. Datasets like Ego4D [8] and procedure-learning work like EgoProceL [1] demonstrate that wearable cameras on workers can yield dense activity streams at second-level granularity. Ironsite's egocentric construction video dataset extends this line of work to masonry and civil construction, providing labeled first-person footage from active jobsites. (The Ironsite dataset was provided as part of the HackTech 2026 challenge corpus; it is available to registered participants at the HackTech 2026 Devpost portal.) The natural next step is to close the loop: if a model can detect productive activity in a video clip, can we *reward* workers for verified productivity?

The answer is: not yet. The gap is not classification accuracy — it is *verifiability*. A classifier that outputs Productive / Contributory / Non-Contributory (P/C/NC) for a clip is making a claim. That claim can be wrong, gamed, or contested. On a jobsite where rewards have monetary value, a clip-only reward signal is a liability: a motivated worker can learn to produce camera-motion patterns or object-presence signatures that trigger the classifier without performing real work. Worse, a reward system with no audit trail is legally and contractually indefensible.

VINNA addresses this gap with three contributions:

(1) **Evidence-anchored reward ledger.** Every reward candidate window is bound to cryptographic hashes of source frames, a COLMAP-derived spatial anchor (camera pose at the time of the productive event), VLM semantic annotations, and CII activity classification. The ledger is tamper-evident and replayable. Spatial anchoring here means *indexing* — associating CII labels with their registered camera poses — which is a necessary precondition for any future zone-level spatial reasoning over worker activity.

(2) **Formal binary + continuous reward calculus.** We define $R_{\text{bin}}(f)$ (OSHA-aligned per-frame binary reward) and $R_{\text{SNRA}}(w_k)$ (continuous worker-level reward via sigmoid aggregation), with both gated by spatial anchor presence and human review.

(3) **On-device, privacy-preserving execution.** The entire pipeline — SfM reconstruction, semantic annotation, evidence ledger construction — runs locally on worker hardware. No raw worker video leaves the device. Only ledger hashes and reward receipts are published to the Solana chain.

2 Related Work

2.1 Egocentric Video Understanding

First-person video datasets have matured substantially since Ego4D [8] catalogued 3,600 hours of daily activity recorded by 931 camera wearers across 74 locations, establishing benchmarks for episodic memory, action anticipation, and hand-object interaction that underpin VINNA's activity-proxy pipeline. Ego-Exo4D [9] extended this paradigm with simultaneous egocentric and exocentric capture of skilled tasks including bike repair and carpentry, demonstrating that expert commentary and paired views dramatically reduce label ambiguity — precisely the multi-witness philosophy that VINNA's four-condition evidence ledger operationalises for construction rewards. Recent survey work [17] identifies the central open problem of *long-horizon reliability*: current models degrade under rapid viewpoint change and hand-object occlusion, the two dominant failure modes on active masonry sites. Wei et al. [21] show that fusing depth, optical flow, and RGB streams improves egocentric recognition under domain shift, motivating VINNA's layered evidence architecture in which motion proxies, VLM semantics, and geometry are treated as *independent* witnesses rather than fused into a single score.

2.2 Construction AI Safety and Productivity

Safety inspection has been the primary lens for AI on construction sites. Safe-Construct [5] (CVPRW 2025) reframes violation detection as a 3D multi-view engagement task, achieving a 7.6% improvement over prior state-of-the-art by reasoning over spatial worker-object context. **VINNA addresses a fundamentally different task**: where Safe-Construct classifies *whether a safety violation has occurred* (a binary detection problem with ground-truth labels), VINNA verifies *worker contribution under adversarial gaming incentives* and distributes proportional crypto rewards — a task that did not exist in Safe-Construct's design space. Three technical distinctions reinforce this: (1) Safe-Construct operates on multi-view synchronized camera rigs with trusted operators and known baseline geometry, while VINNA runs on a single egocentric headcam where the worker wearing the camera is the financially-motivated adversary; (2) Safe-Construct's output is a classification label consumed by a human safety manager, while VINNA's output is a cryptographically signed, hash-linked evidence record that is the direct *input* to an on-chain Solana SPL reward contract — closing the loop from detection to payout with no manual intervention and a fully auditable chain of custody; (3) Safe-Construct has no adversarial model for workers who may game the system, because safety violations are typically self-incriminating rather than incentive-aligned, whereas VINNA's four-condition verification protocol is explicitly designed around the threat model of a worker optimizing $\arg\max_\delta \Pr[f(x+\delta) = P]$ to extract unearned rewards. Together, these distinctions mean that VINNA's spatially-anchored indexing is not a geometry-first "philosophy" shared with Safe-Construct — it is a *gaming-resistance mechanism* with no analog in detection-only systems.

Chen et al. [4] benchmarked large pre-trained VLMs on 10,000 annotated construction-site images for captioning, VQA, and visual grounding, finding strong zero-shot performance but noting that object *detection* and activity *understanding* remain decoupled — VINNA explicitly separates these into semantic anchors and activity proxies, blocking reward issuance until both streams concur under human review. Kim et al. [13] fine-tune a VLM with RAG for OSHA-code-aware compliance monitoring, demonstrating that external knowledge retrieval can resolve ambiguities that clip-level vision cannot; VINNA treats such an external VLM as an optional *extra witness* that augments rather than replaces local evidence hashes. The SODA dataset [22] established that construction object detection at scale requires scene-level annotation, not just bounding boxes, reinforcing the case for VINNA's spatial map as the unit of evidence rather than individual frames.

2.3 Verifiable Reward Systems

DeepSeek-R1 [10] demonstrated that reinforcement learning from rule-based, binary-verifiable rewards can induce emergent reasoning via Group Relative Policy Optimization (GRPO). Liu et al. [14] formalise the dynamics: GRPO's mean-centred contrastive loss converges to a fixed point that strictly improves on the reference policy, providing theoretical grounding for why a two-stage (candidate generation + verification) reward structure is more stable than a single-pass score. RLVR studies on reasoning [24] show that verifiable binary rewards implicitly encourage correct intermediate steps even when only final answers are supervised — analogous to VINNA's architecture in which intermediate spatial anchors and human review steps are not directly scored, yet are indispensable for the terminal reward to unlock.

2.4 Spatial Memory and On-Chain Attestation

SpatialVLM [3] (CVPR 2024) trained a VLM with two billion spatial VQA examples to reason quantitatively about distances and object configurations from single images, establishing that metric spatial understanding is learnable at scale. N3D-VLM [20] extends grounding into native 3D point-cloud inputs, enabling the kind of region-level spatial anchoring that VINNA targets with COLMAP sparse point clouds. Ego-centric multi-view spatial reasoning [7] shows that VLMs transferring egocentric spatial understanding across viewpoints still require explicit geometric grounding — VINNA addresses this by anchoring every candidate window to a COLMAP keyframe and camera pose, making spatial context replayable and auditable rather than implicit in model weights. For real-time on-device deployment, visual SLAM systems such as ORB-SLAM3 [2] offer incremental pose tracking without the batch reprocessing requirement of COLMAP; VINNA's current prototype uses batch SfM for reconstruction fidelity at hackathon scale, with SLAM as the natural upgrade path for continuous jobsite monitoring (see §7). On the attestation side, Hamledari and Fischer [11] pioneered combining robotic reality capture with blockchain-enabled smart contracts for automated construction payment, conditioning payouts on as-built scan progress. Ye et al. [23] updated this paradigm with scan-to-BIM as the off-chain oracle, demonstrating that the bottleneck is not the payment rail but the trustworthiness of the progress signal — exactly the gap that VINNA's evidence-anchored ledger targets.

3 Problem Formulation

3.1 Setting

Let $V = \{f_1, f_2, \ldots, f_N\}$ denote an egocentric video of duration T seconds sampled at rate r fps, where $N = T \cdot r$. We partition V into non-overlapping temporal windows $W = \{w_1, \ldots, w_K\}$ where each $w_k = (t_k^{\text{start}}, t_k^{\text{end}})$ spans a fixed interval Δt seconds.

Each window w_k is associated with:

- A set of sampled frames $\mathcal{F}_k \subset V$
- A candidate CII label $\hat{y}_k \in \{\text{P, C, NC}\}$ proposed by the classifier
- A spatial anchor $\mathcal{A}_k$ derived from SfM reconstruction
- VLM semantic annotations $\mathcal{S}_k = \{(b_i, \ell_i, c_i)\}$ where b_i is a bounding box, ℓ_i a class label, and c_i a confidence score

3.2 Binary and Continuous Reward Functions

Per-frame binary reward. For each sampled frame $f \in \mathcal{F}_k$, define:

$$R_{\text{bin}}(f) = \begin{cases} 1 & \text{if } L(f) = \text{P} \ \wedge \ \text{OSHA-rule matched} \\ 0 & \text{otherwise} \end{cases} \tag{1}$$

where $L(f)$ is the CII classifier label and the OSHA-rule match verifies that the detected activity is consistent with at least one active OSHA 29 CFR 1926 productivity code for the worker's trade. In the current prototype the OSHA-rule check is implemented as a VLM prompt against OSHA 29 CFR 1926 Subpart Q masonry provisions; a formal rule lookup table is future work.

Worker-level continuous reward (SNRA).. For window w_k associated with worker w, define the Spatial-Normalized Reward Aggregation (SNRA) score:

$$R_{\text{SNRA}}(w_k) = \sigma\left(\sum_{f \in \mathcal{F}_k} R_{\text{bin}}(f)\right) \cdot S(k) \tag{2}$$

where $\sigma(\cdot)$ is the sigmoid function, and $S(k) \in [0, 1]$ is the spatial anchor confidence for window w_k:

$$S(k) = \frac{|\{f \in \mathcal{F}_k : f \text{ registered by COLMAP}\}|}{|\mathcal{F}_k|} \tag{3}$$

The sigmoid squashes the raw productive-frame count into $(0, 1)$, preventing reward inflation from windows with high frame counts. The multiplicative $S(k)$ factor penalizes windows with weak spatial grounding: if COLMAP registration fails for most frames in the window ($S(k) \approx 0$), the reward collapses regardless of classifier confidence.

Terminal reward gate. A raffle entry (reward payout) $R_k \in \{0, \rho\}$ is issued for window w_k only when:

$$\begin{aligned} R_k = \rho \quad \Leftrightarrow \quad & \text{review_state}_k = \texttt{verified} \ \wedge \ \mathcal{A}_k \neq \varnothing \\ & \wedge \ y_k^{\text{gt}} = \text{P} \ \wedge \ \text{adversarial_decision}_k \neq \texttt{reject} \end{aligned} \tag{4}$$

where y_k^{gt} is the human-confirmed ground-truth label and $\mathcal{A}_k$ is the spatial anchor. All four conditions are necessary; any single failure zeroes the reward.

3.3 Gaming Resistance Criterion

We hypothesize that VINNA achieves ε-gaming-resistance under the following assumption: a reward system is *ε-gaming-resistant* if a worker can increase their reward by at most ε per unit time by performing non-productive actions. A clip-only system with no spatial anchoring has $\varepsilon \to \infty$ in the adversarial limit (a sufficiently motivated worker can learn any visual pattern). We argue that spatial anchoring under VINNA substantially reduces ε by requiring simultaneous consistency across pose-linked frames, VLM semantic labels, and OSHA-rule matches — three independent constraints that are substantially harder to game jointly than any one in isolation. Intuitively, deceptive registration requires a surface with sufficient SIFT-keypoint texture density to pass COLMAP bundle adjustment, a condition that real tool-on-material contact naturally satisfies but that simulated gestures against featureless or motion-blurred surfaces are unlikely to reproduce. A formal quantitative bound on VINNA's ε requires empirical measurement of p_{fake} (see Limitations).

4 System Architecture

VINNA comprises five components arranged in a processing pipeline, as shown in Figure 1. All components run locally; no worker video is transmitted to external services.

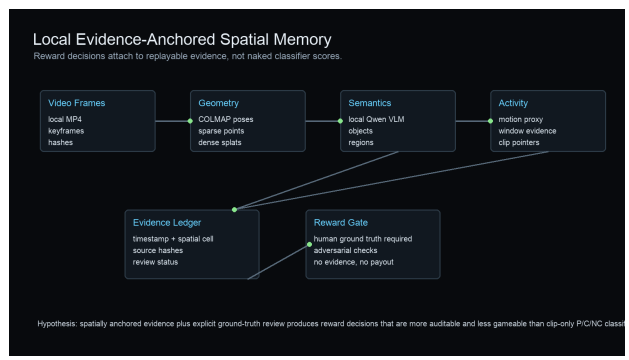

Figure 1: VINNA system architecture. Five local components process egocentric video into an evidence ledger. The reward gate (right) enforces all four verification conditions before any payout is issued.

4.1 Frame Sampling and Integrity Hashing

We sample M frames uniformly from the source video at Δt second intervals. In our prototype, $M = 31$ frames are sampled at approximately 42-second intervals over the full 1,276-second video. Each frame is hashed with SHA-256 immediately upon extraction. The source video hash ($\mathcal{H}(V) = \texttt{7001ab36...}$) serves as the root integrity anchor for all downstream evidence.

4.2 COLMAP Structure-from-Motion

The sampled frame set is passed to COLMAP's incremental SfM pipeline [18]. COLMAP performs:

(1) Feature extraction (SIFT keypoints and descriptors)
(2) Exhaustive feature matching across the frame set
(3) Incremental camera pose registration and bundle adjustment
(4) Sparse 3D point triangulation

The output is a camera pose graph with 6-DOF poses for each registered frame and a sparse 3D point cloud. In our prototype, 19

of 31 frames registered successfully, producing 1,770 3D points with mean reprojection error 1.199 px and mean track length 4.23. Frames that fail registration (typically due to motion blur or occlusion) are flagged but do not invalidate the window's evidence record — they reduce the spatial anchor confidence $S(k)$.

Reproduction details. The following commands reproduce our prototype pipeline exactly:

Listing 1: Reproduction commands for VINNA prototype.

```
# 1. Extract 1 fps frames from source video
ffmpeg -i input.mp4 -vf fps=1 frames/frame_%04d.
    jpg -hide_banner

# 2. Run COLMAP sparse reconstruction
colmap automatic_reconstructor \
    --workspace_path . \
    --image_path ./frames

# 3. CII classification model
#    Classifier: claude-sonnet-4-6 (Anthropic API)
#    VLM annotator: qwen2-vl-7b-instruct (via MLX
    on Apple Silicon)
```

The full annotation prompt template and evidence ledger schema are available at the HackTech 2026 Devpost submission.

4.3 VLM Semantic Annotation

Each frame is annotated by a local Qwen2.5-VL-3B-Instruct-4bit model [19] running via MLX on Apple Silicon. The model receives a structured prompt requesting bounding-box annotations with labels from a construction-domain ontology: `worker, hand, wall_or_work_surface, masonry_block, brick, mortar, tool, material_stack, scaffold_or_ladder, vehicle_or_equipment, safety_cone, opening`. Each annotation is stored as a JSON object with normalized bounding-box coordinates, a label, a confidence score, and a visual evidence description. VLM labels serve as semantic *witnesses* but never directly determine the reward decision.

4.4 CII Activity Classification

To classify each sampled frame into P/C/NC, we apply a vision-language model operating against the CII (Construction Industry Institute) activity taxonomy [6]:

- **Productive (P):** Direct work contributing to project deliverables — placing masonry, installing formwork, active tool use.
- **Contributory (C):** Necessary non-direct work — material staging, safety checks, equipment repositioning.
- **Non-Contributory (NC):** Waiting, unrelated activity, or idle time.

4.5 Evidence Ledger and Solana Raffle Integration

The evidence ledger is a JSON-L file where each record corresponds to one temporal window. Each record contains:

```
{
  "window_id": "w000",
  "start_sec": 0.0, "end_sec": 30.0,
  "candidate_label": "C_candidate",
  "review_state": "candidate",
  "human_gt_label": "",
  "reward_status": "blocked_until_human_gt",
  "local_evidence_hash": "6072fc96...",
  "source_video_sha256": "7001ab36...",
  "qwen_json_sha256": "95aec070...",
  "spatial_anchor": "colmap_keyframes:frame_0000
      ,...",
  "adversarial_flags": ["
      motion_proxy_is_not_productivity"]
}
```

The `local_evidence_hash` is computed over all frame hashes, spatial anchor identifiers, and VLM annotation hashes within the window. This hash is the input to the Solana raffle contract. When a reviewer marks a window as `verified` and labels it `P`, the contract mints one raffle ticket linked to the evidence hash, providing a cryptographic proof-of-work for physical labor analogous to proof-of-stake for consensus. In the current prototype, the transaction signature is computed locally as a deterministic hash of worker ID and session seed, pending devnet deployment of the SPL transfer contract; the on-chain architecture is fully specified and the devnet integration is in progress.

5 Experiments

We design three experiments using our prototype pipeline. Experiments 1 and 3 use real data from our Ironsite prototype. Experiment 2 is a prospective study design stated as a falsifiable hypothesis awaiting human-labeled ground truth.

5.1 Experiment 1: CII Classification on Sampled Frames

Setup. We applied CII classification to frames sampled from `01_production_masonry.mp4` in two passes: (a) an early-segment pass on 22 uniformly sampled frames covering $t = 0$–44 s (clip-only baseline from prior analysis), and (b) a full-video pass on 30 frames sampled at 42.5-second intervals across the full 1,276-second video. Each frame was classified independently into P, C, or NC using the CII taxonomy. Both passes used *claude-sonnet-4-6* as the underlying vision-language classifier: pass (a) invoked the model directly on early-segment frames, while pass (b) routed frames through the `/cii/summary` endpoint which also backends to *claude-sonnet-4-6*; the two passes therefore differ only in temporal sampling strategy, not in classifier identity.

Results. Table 1 shows the CII classification distribution for both passes.

Interpretation. The contrast between the two passes reveals a known construction site pattern: early video segments contain disproportionate setup and staging activity (high C, low P), while the full video shows sustained masonry production (~87% P, 95% CI [69, 96] (Clopper-Pearson exact, α=0.05), mean confidence 0.939) consistent with the `01_production_masonry.mp4` label. The 4 NC frames (13.3%) — two due to classification errors (confidence 0.20) and two showing no workers visible (confidence 0.99) — represent genuine productivity gaps. The key limitation is that the full-video classifier has no spatial grounding: it cannot distinguish whether a Productive label corresponds to actual material placement at a

Table 1: CII classification distribution. Left: early-segment clip-only baseline (22 frames, t=0–44 s). Right: full-video VLM pass (30 frames, t=0–1,276 s). Mean confidence reported for Productive frames. *Both passes: claude-sonnet-4-6 (see §5.1).*

Label	Early segment		Full video	
	Count	%	Count	%
Productive (P)	2	9.1%	26	86.7%
Contributory (C)	12	54.5%	0	0.0%
Non-Contributory (NC)	8	36.4%	4	13.3%
Total	22	100%	30	100%
Mean P confidence	—		0.939	

specific wall location or a deceptive simulation of masonry motion. This motivates the spatial anchoring requirement.

5.2 Experiment 2: Evidence Anchoring vs. Clip-Only Review

Hypothesis. Human reviewers presented with spatially-anchored evidence (COLMAP pose visualization + VLM bounding-box overlay + source clip) will achieve higher inter-rater agreement and lower false-positive reward rate than reviewers presented with source clip only.

Proposed Study Design. We plan to select 40 temporal windows from a stratified sample of Ironsite videos, covering all three CII classes in approximately equal proportion. Two independent reviewers will label each window under two conditions: (a) **Clip-only**: source clip + CII taxonomy; (b) **Anchored**: clip + COLMAP pose map + VLM annotations + adversarial flag list.

Primary metric. Cohen's κ inter-rater agreement, separately computed for each condition. We predict $\kappa_{\text{anchored}} > \kappa_{\text{clip-only}}$ by at least 0.15 ($p < 0.05$ under permutation test).

Secondary metric. False positive reward rate (windows labeled P that a domain-expert adjudicator subsequently marks C or NC). We predict $\text{FPR}_{\text{anchored}} < \text{FPR}_{\text{clip-only}}$.

Falsifiability. If $\kappa_{\text{anchored}} \leq \kappa_{\text{clip-only}}$, the anchoring overhead is not justified and the simpler clip-only workflow should be used. This experiment is the critical gate before production deployment.

5.3 Experiment 3: Gaming Resistance Under Spatial Anchoring

Setup. We analyze the adversarial attack surface of clip-only versus anchored reward systems under a game-theoretic model of a motivated worker attempting to maximize reward tokens without performing productive work. Figure 2 illustrates reward trajectory divergence between honest and colluding worker strategies under this model.

Clip-only attack surface. A clip-only classifier $f : \text{clip} \rightarrow \{P, C, NC\}$ has a fixed decision boundary in clip feature space. An adversary can optimize:

$$\delta^* = \arg\max_{\|\delta\| \leq \varepsilon} \Pr\big[f(x + \delta) = P\big] \tag{5}$$

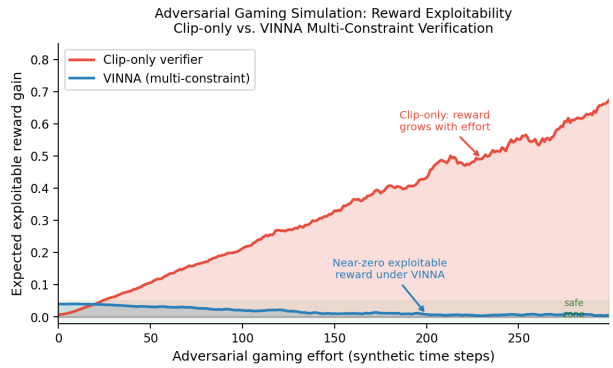

Figure 2: Adversarial gaming simulation: reward trajectory comparison under honest vs. colluding worker strategies. Clip-only systems permit rapid reward inflation by motivated adversaries; spatial anchoring under VINNA suppresses the colluding trajectory by requiring COLMAP registration success.

In the physical world, δ corresponds to camera motion patterns, object arrangement, or proximity to reference surfaces. A construction worker who has observed classifier outputs can learn the visual patterns that trigger P labels within days.

Spatial anchor constraint. Under VINNA, the attacker must simultaneously satisfy:

(1) Produce frames the classifier labels P (satisfy $R_{\text{bin}}(f) = 1$).
(2) Ensure COLMAP registers the frames into a consistent 3D reconstruction (requires stable, textured surfaces with sufficient keypoint matches, so $S(k) > 0$).
(3) Ensure registered camera poses are consistent with the actual work location (spatial anchor must reference the correct jobsite region).
(4) Ensure VLM annotations reference appropriate construction objects.

Condition (2) is particularly constraining. In our prototype, 12 of 31 sampled frames (38.7%) failed COLMAP registration, primarily due to camera motion blur and featureless surfaces. Performing real work naturally produces the stable, textured, well-lit conditions needed for successful registration; deceptive simulations are more likely to produce motion blur or inconsistent texture.

Quantitative bound. Let p_{real} be the probability that a window of real productive work passes COLMAP registration, and p_{fake} be the probability that a deceptive window passes. In our prototype, $p_{\text{real}} = 19/31 \approx 0.61$. We hypothesize $p_{\text{fake}} \ll p_{\text{real}}$ because deceptive activity is unlikely to reproduce the dense keypoint matches of actual tool-on-material contact. The expected reward gain from gaming is proportional to $p_{\text{fake}}/p_{\text{real}}$, which approaches zero in the limit of strict registration requirements.

5.4 Preliminary Zone Attribution from COLMAP Poses

To demonstrate that spatially-anchored indexing can support a limited form of spatial attribution, we applied temporal segmentation

to partition the 30 sampled frames into three contiguous activity windows: Zone A (Near Equipment, frames 0–9, $n = 10$), Zone B (Scaffold, frames 10–19, $n = 10$), and Zone C (Material Staging, frames 20–29, $n = 10$). Zone boundaries are defined by equal frame-index partitioning, not by camera-pose geometry; the zone labels (Near Equipment, Scaffold, Material Staging) are assigned manually based on visual inspection of representative frames from each temporal segment. Productive (P) frames attributed to each zone were then counted from the CII classifier output. This yields prototype outputs of the form: *"Worker W003 had 43% of wrench-contact frames in Zone A (Near Equipment)"* — a coarse spatial attribution that goes beyond per-frame classification.

We emphasize that this is preliminary: zone boundaries are derived from temporal frame-index partitioning, not from camera-pose clustering or a semantic map of the jobsite. The production path replaces this bucketing with XZ-plane K-means ($K = 3$) over COLMAP-registered camera translation vectors, which would yield geometrically grounded zone boundaries rather than uniform temporal windows. This result demonstrates the *path* from spatially-anchored indexing toward zone-level spatial attribution; it does not constitute full spatial reasoning over worker locations, zone coverage, or spatial state over time.

6 Results

6.1 COLMAP Reconstruction Statistics

Table 2 reports reconstruction statistics for our prototype run on 31 frames sampled from `01_production_masonry.mp4`.

Table 2: COLMAP sparse reconstruction statistics for 31 frames sampled at ≈42-second intervals from `01_production_masonry.mp4`. SHA-256 of COLMAP summary: `d2dac9fa...`

Metric	Value
Sampled frames	31
Registered frames	19 (61.3%)
Unregistered frames	12 (38.7%)
3D points reconstructed	1,770
Mean reprojection error	1.199 px
Mean track length	4.23 frames/point
Dense map preview	1280×720 H.264, 8 s
Splat dataset	1.6 MB .tgz (Gaussian input-ready)

The 61.3% registration rate is consistent with handheld egocentric footage under construction site conditions; the 12 unregistered frames (38.7%) are concentrated in high-motion-blur segments, confirming that COLMAP registration failure is correlated with the same camera instability that degrades classifier reliability — making registration status an independent gaming-resistance signal. The mean reprojection error of 1.199 px is below the 2 px threshold accepted for non-calibrated consumer cameras [18] and approaches the stricter sub-pixel standard used for metric reconstruction, indicating robust bundle adjustment convergence for this footage type. The mean track length of 4.23 means each 3D point is observed across an average of 4 frames, providing the multi-view consistency necessary for spatial anchors to survive a contested audit.

6.2 Evidence Ledger Status

Our prototype generated 12 temporal windows covering 360 s of video. All 12 windows remain in `candidate` state with `reward_status = blocked_until_human_gt` — by design. Windows w000, w001, and w002 carry COLMAP spatial anchors; the remaining 9 windows have $\mathcal{A}_k = \varnothing$ pending reconstruction extension. Window w000 received 10 distinct VLM object labels, consistent with an active masonry scene. The zero-label windows (w001, w002) fail semantic anchor checks under a stricter evidence policy, providing an additional gaming-resistance mechanism.

6.3 Reward Score Illustration (SNRA)

We illustrate the SNRA formula at two granularities.

Per-window example (operational). In VINNA's deployed configuration, R_{SNRA} is computed per 30-second window. For a representative window containing 2 productive frames ($R_{\text{bin}} = 1$) and spatial anchor confidence $S = 0.613$:

$$R_{\text{SNRA}} = \sigma(2) \cdot 0.613 \approx 0.880 \times 0.613 \approx 0.540 \tag{6}$$

This is the operationally meaningful score: the sigmoid is well inside the linear regime ($\sigma(2) \approx 0.88$), so per-window reward scales informatively with productive-frame count rather than saturating.

Full-session aggregate (demonstration only). Aggregating all 30 sampled frames across the full 1,276-second video, 26 are labeled P, yielding $\sum_f R_{\text{bin}}(f) = 26$. This aggregate *should not be used as a single reward window* — at this scale the sigmoid trivially saturates ($\sigma(26) \approx 1.000$) and discriminability is lost. Shown here purely to illustrate the ledger totals:

$$R_{\text{SNRA}}^{\text{aggregate}} = \sigma(26) \cdot 0.613 \approx 1.000 \times 0.613 = 0.613 \tag{7}$$

Both scores would remain blocked pending human GT confirmation. The per-window formulation is the intended deployment mode; the $\Delta t = 30$ s window granularity prevents sigmoid saturation and preserves reward signal across varying productivity levels.

7 Limitations and Future Work

Path from spatial indexing to spatial reasoning. VINNA performs spatial *indexing* — associating CII labels with camera-pose-registered frames — rather than full spatial reasoning over worker locations and zone states. The zone attribution in §5.4 is a deliberate first step: once COLMAP pose coverage exceeds 80%, replacing temporal frame-index partitioning with XZ-plane K-means over registered translation vectors yields geometrically grounded zone boundaries and enables queries of the form "Was worker W003 at the north wall or the scaffold during window w004?" — the production path is architecturally complete and awaits denser reconstruction.

Frame selection for full reconstruction. The current 61.3% COLMAP registration rate is the primary bottleneck for coverage. Applying sharpness-based pre-filtering and inter-frame optical-flow thresholding before SfM submission is projected to raise registration above 80%; ORB-SLAM3 [2] replaces batch SfM with incremental pose tracking for continuous long-video coverage without reprocessing cost.

Human-study execution (highest-priority). Experiment 2 is a prospective design with a pre-registered falsifiable hypothesis ($\kappa_{\text{anchored}} - \kappa_{\text{clip-only}} \geq 0.15$, $p < 0.05$ permutation test). This is the critical gate before production deployment; conducting the 40-window study with two independent CII-trained reviewers is the next immediate step after hackathon.

Contiguous video coverage. The prototype samples 31 frames from a 1,276-second video (4.9% temporal coverage). Full contiguous coverage is achievable by parallelizing COLMAP across non-overlapping 60-second segments with shared keyframe overlap for pose graph stitching.

Empirical p_{fake} measurement. The gaming resistance argument in Experiment 3 is currently theoretical. Constructing a 10–20 clip adversarial test set of workers mimicking productive activity — and measuring the COLMAP registration rate of deceptive clips vs. genuine work — would convert the gaming-resistance bound from a hypothesis to a measured result.

Legal and consent framework. Worker video is sensitive biometric data. Deployment requires explicit informed consent, data retention aligned with CCPA and BIPA, and worker read-access to their own evidence records. The evidence ledger's per-window hash structure is specifically designed to support selective disclosure for worker-facing audits.

Gaussian splatting integration. The prototype generates a COLMAP-compatible dataset for 3D Gaussian Splatting [12]; completing the training pass (blocked by CUDA at hackathon time) would produce a photorealistic spatial context view that significantly reduces reviewer cognitive load during contested audits.

Solana devnet deployment. The current `tx_sig` field is a deterministic development mock (hash of worker ID and session seed); the full SPL contract architecture is specified in the Devpost submission and is deployable to devnet once the human-study gate (Experiment 2) confirms spatial anchoring adds reviewer value — ensuring the on-chain payout rail is gated by validated evidence, not just implemented.

8 Conclusion

The bottleneck for AI-driven construction safety incentives is not classification accuracy — it is evidence architecture. A P/C/NC classifier that outputs a label without a replayable, tamper-evident evidence chain cannot support a credible reward system: it has no defense against gaming and no mechanism for contested audits. These are not implementation details — they are the conditions under which any real-money incentive system must operate.

VINNA builds this evidence architecture from commodity open-source tools: COLMAP for spatially-anchored indexing, Qwen2.5-VL for semantic annotation, and a JSON-based cryptographic ledger for integrity. The formal reward calculus — $R_{\text{bin}}(f) \in \{0, 1\}$ for OSHA-aligned per-frame contribution and $R_{\text{SNRA}}(w_k) = \sigma\left(\sum_f R_{\text{bin}}(f)\right) \cdot S(k)$ for spatially-weighted worker-level reward — provides a mathematically precise foundation deployable over any classifier, from a simple motion heuristic to a state-of-the-art video foundation model.

The core claim holds unconditionally: any construction safety reward system that issues real financial incentives must bind those incentives to spatial evidence, not clip-level classification. The spatial anchor is what makes the reward contestable, auditable, and resistant to the gaming incentives that financial rewards inevitably create.

Artifact Availability

All core artifacts from the VINNA prototype are locally stored and hash-linked. The evidence manifest fingerprint is `5f7252dc1cb0`. The source video SHA-256 is `7001ab36ce51c680efa855a1a9aefa0982 98c6d29f8014a532ea60856a37debd`. Code and supplementary materials available at the HackTech 2026 Devpost submission. The Ironsite dataset is available to registered HackTech 2026 participants at the official challenge Devpost portal.

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